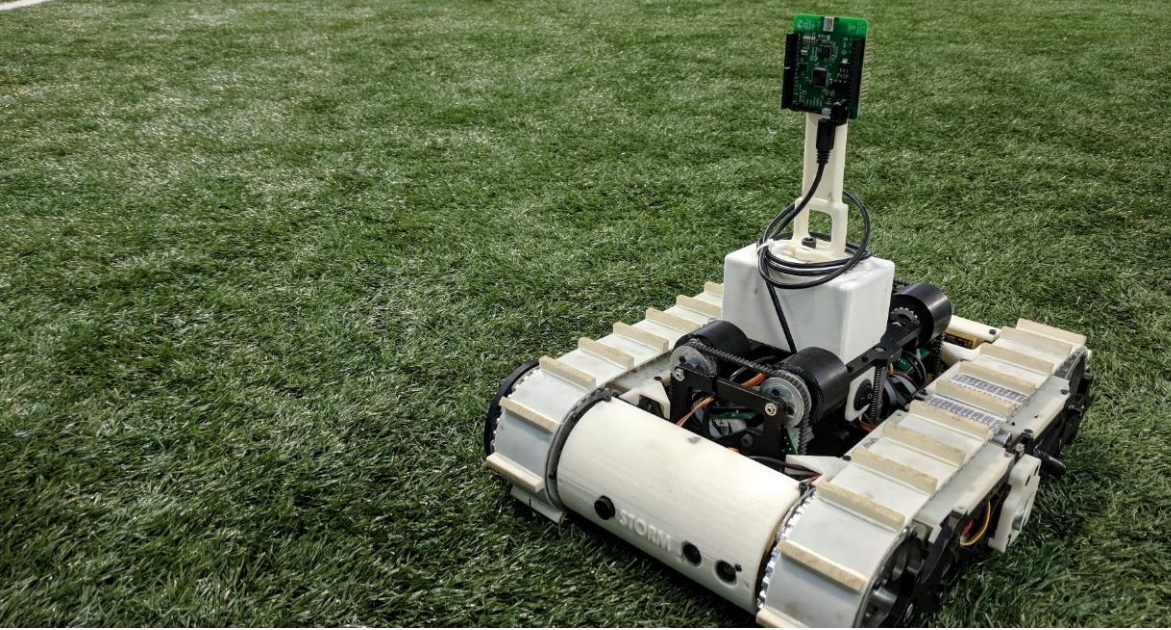




Path Planning for Pioneer P3-AT

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PROJECT DEFINITION

- **Problem:** Find an efficient path for a Pioneer P3-AT robot to navigate from its initial location to a goal in a dynamic environment
- The motion planning execution must satisfy the constraints of skid-steer wheeled robot dynamics
- This includes finding a path to the goal and maneuvering the vehicle along this path to avoid obstacles
- Maps will be given, but will change over time
- Algorithm must be adaptive to changes, without having to replan from scratch

We make various assumptions like we have perfect sensing, access to the global map at all times, the robot will not run out of battery, will get its global coordinates, moving obstacles have linear trajectories, etc, in order to focus on the motion planning aspects of this problem.

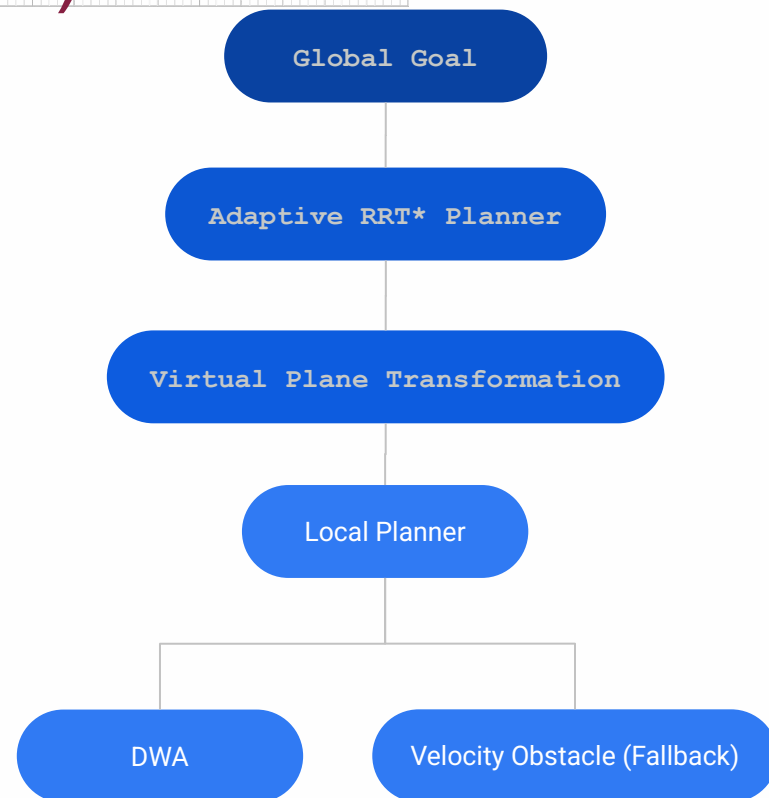


METHODOLOGY (As of now)

Hybrid Global-Local Approach

1. Global Planning
 - Adaptive RRT* with 3D→4D switching near obstacles
 - Virtual Plane obstacle projection
2. Local Adaptation
 - Dynamic Window Approach (DWA) with PID-controlled collision cones
 - Velocity Obstacle fallback for emergencies <1m

Reference: F. Belkhouche, "Reactive Path Planning in a Dynamic Environment," in IEEE Transactions on Robotics, vol. 25, no. 4, pp. 902-911, Aug. 2009, doi: 10.1109/TRO.2009.2022441.



DELIVERABLES

Path Planning in Dynamic Environments (Simulation)

- Implement adaptive motion planning for a **Pioneer P3-AT** robot in **Gazebo**.
- Navigate through multiple simulated environments with varying layouts, static and dynamic obstacles with **different behaviours/trajectories**.

Performance Analysis & Optimization

- Evaluate **path efficiency**, **optimality**, and **adaptability** in dynamic scenarios.
- Compare different path-planning algorithms and refine for better real-time performance.

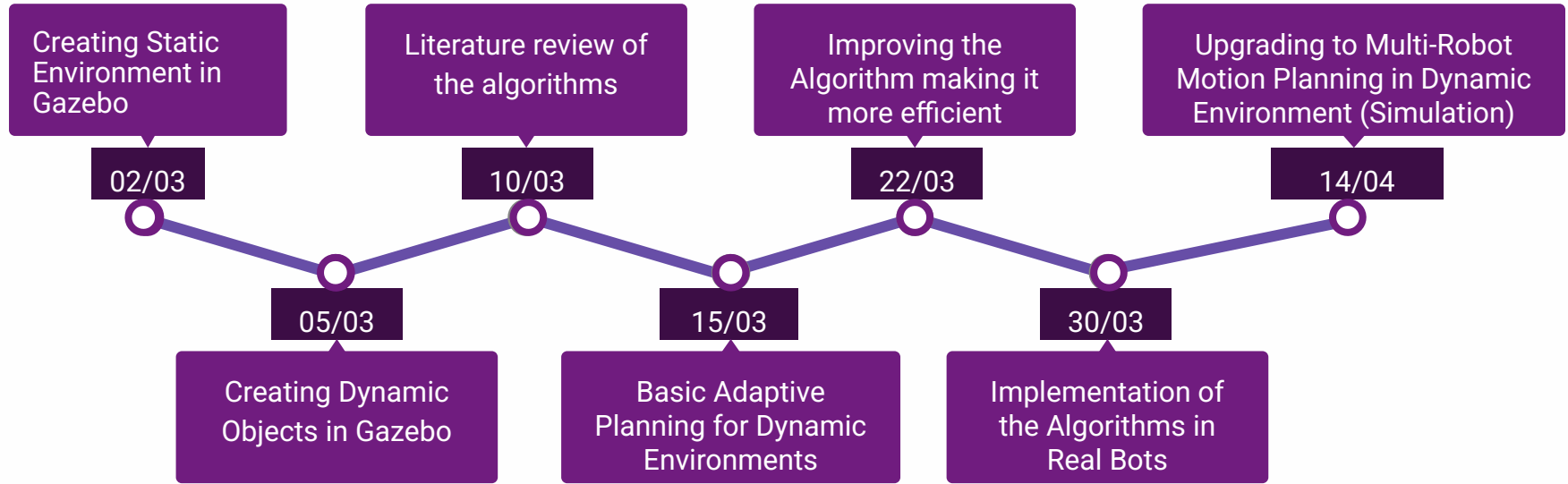
Real-World Deployment

- Validate the most optimal and efficient algorithm on a **physical robot**.
- Optimize for real-world uncertainties like sensor noise/drift and wheel slippage.

Extended deliverables:

- Extend the simulation framework to handle **multiple autonomous robots**, each with different goal positions.
- Implement a decentralized/cooperative planning strategy to **avoid collisions and optimize efficiency**.

MILESTONES AND TIMELINE



QUESTIONS!

- Is local planning allowed?
- What metrics are recommended to tune the PID controller collision cones?
- What existing D-Lite/RTAB-Map implementations should we benchmark against for dynamic adaptability?
- Is path length or collision-free execution more critical for evaluation—how would you weight these in scoring?
- How do we adapt Velocity Obstacle to prevent reciprocity deadlocks without centralized control?